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Introduction

This internal report is addressed to the KPMG engagement manager responsible for GreenTech Solutions Ltd. (GTS). It evaluates how KPMG should approach the concurrent financial statement audit and first-time limited assurance engagement over GTS's selected Environmental, Social and Governance (ESG) indicators. The analysis is based on the GTS case facts, relevant International Standards on Auditing (ISAs), International Standard on Assurance Engagements (ISAE) 3000 and ISAE 3410, and the potential use of artificial intelligence (AI) and blockchain. It focuses on assurance boundaries, three ESG evidence challenges—data quality, measurement inconsistency and supply-chain reliance—stakeholder-sensitive materiality, and technology-enabled procedures that improve assurance quality without weakening professional judgement.

Section 1: The Auditor's Evolving Responsibility in Auditing ESG Reports

Because the financial audit and ESG assurance run concurrently, KPMG should first define each engagement's subject matter, criteria, evidence requirements and reporting form (International Auditing and Assurance Standards Board [IAASB], 2013).

1.1 The Assurance Engagement

Both engagements enhance users' confidence, but at different levels. The financial statement audit is conducted under the ISAs and covers GTS's financial statements as a whole. By contrast, the ESG engagement is performed under ISAE 3000 and covers selected indicators only: recycled content percentage, Scope 1 and 2 greenhouse gas (GHG) emissions and workforce diversity metrics. Accordingly, they differ in subject matter, assurance level, procedures and reporting outcome (IAASB, 2013).

Table 1. Financial Audit versus ESG Limited Assurance for GTS

Area	Financial statement audit	ESG limited assurance
Standards	ISAs	ISAE 3000; ISAE 3410 for GHG emissions
Subject matter	Financial statements as a whole	Selected ESG indicators
Assurance level	Reasonable / high	Limited / moderate
Main procedures	Risk assessment, controls and substantive testing	Inquiry, analytics and limited inspection
Reporting outcome	Positive audit opinion	Negative assurance conclusion

As shown in Table 1, the financial audit requires reasonable assurance that the financial statements are free from material misstatement and sufficient appropriate evidence to support a positive opinion (IAASB, 2009a; IAASB, 2009b). By contrast, the ESG engagement is governed by ISAE 3000, with ISAE 3410 applicable to Scope 1 and 2 GHG emissions (IAASB, 2012; IAASB, 2013). It is narrower in subject matter because it does not cover GTS's overall sustainability performance.

The assurance level is the key distinction. Reasonable assurance is high but not absolute, while limited assurance provides a lower level of confidence and is normally supported by less extensive procedures (IAASB, 2013). For GTS, limited assurance remains appropriate because the engagement terms specify selected metrics, this is the company's first ESG assurance engagement, and the relevant data systems and external evidence are less mature. The greenwashing allegations do not automatically change the agreed assurance level; instead, they increase the need for sceptical risk assessment and targeted procedures within the limited-assurance scope (IAASB, 2021).

The reporting language should make this boundary clear. For GTS, KPMG's conclusion could state that "nothing has come to our attention that causes us to believe that GTS's recycled content percentage, Scope 1 and 2 GHG emissions and workforce diversity metrics are not prepared, in all material respects, in accordance with the stated reporting criteria." This negative-form conclusion is consistent with limited assurance, but it is not equivalent to audit-level assurance (IAASB, 2013). Users should not read it as assurance over the whole Sustainability Report, Supplier X's ethical sourcing, Scope 3 customer-use claims or all green marketing statements. This reduces the expectation gap and keeps KPMG's conclusion appropriately bounded.

1.2 Challenges and Professional Scepticism

GTS's ESG assurance engagement has high inherent risk because the selected indicators are non-financial, estimation-heavy and supported by less mature controls than financial information. The key issue is not only whether figures can be recalculated, but whether the data sources, reporting criteria and evidence trail are reliable enough to support a limited assurance conclusion (IAASB, 2021). Table 2 summarises the three case-specific ESG assurance challenges and the related sceptical audit responses.

Table 2. ESG Assurance Challenges and Sceptical Responses for GTS

Challenge	GTS trigger	Sceptical audit response
Data quality and granularity	Marketing / CSR data; internal AI carbon model	Test controls, inputs, assumptions and model access
Measurement and consistency	Recycled content percentage	Benchmark criteria and test consistent application
Third-party and supply-chain reliance	Supplier X; email surveys; NGO allegations	Corroborate externally and document access limits

Note: CSR = corporate social responsibility; NGO = non-governmental organisation.

Data Quality and Granularity

Since Marketing / CSR manages GTS's sustainability data with limited finance or internal audit involvement, documentation, review and reputational-bias risks increase. The internally developed AI model for product-level Scope 3 carbon footprints adds estimation risk because Scope 3 data depends on value-chain information that is difficult to measure and corroborate (Greenhouse Gas Protocol, 2011; Association of Chartered Certified Accountants [ACCA], 2023). KPMG should seek access through the engagement terms and request model documentation, data sources, key assumptions, change logs, validation results and management review evidence. If access to proprietary parameters or training data is restricted, KPMG should document the limitation, perform sensitivity or benchmark testing, escalate unresolved restrictions, and assess the effect on the assurance conclusion.

Measurement and Consistency

The recycled content allegation makes the definition of "recycled content" a central audit issue. Even if the percentage is arithmetically accurate, it may be misleading if the calculation basis is unclear or inconsistently applied. Global Reporting Initiative (GRI) 301 provides a relevant reference for materials and recycled input materials, but GTS still needs a clear method suitable for its products (Global Sustainability Standards Board [GSSB], 2016a; IAASB, 2013). KPMG should compare GTS's definition with GRI 301, supplier certificates, industry practice and prior-period methodology, and test whether the same basis, such as weight, cost or product

category, is applied consistently across product lines and periods.

Third-Party and Supply Chain Reliance

Supplier X creates a further evidence challenge because labour and emissions information sits outside GTS's direct control. Supplier data collected through sporadic email surveys is weak evidence because it is self-reported, informal and difficult to verify independently. Responsible supply-chain due diligence requires companies to identify, prevent and address adverse impacts in business relationships, so KPMG should not rely on supplier self-reporting alone (Organisation for Economic Co-operation and Development [OECD], 2018). The team should seek corroboration from contracts, purchase records, independent certifications, inspection reports, emissions documents and credible external sources. If site records or direct supplier access are unavailable, KPMG should record the evidence gap, management's explanation, alternative procedures and unresolved contradictions, because audit evidence must be sufficient and appropriate rather than merely available (IAASB, 2009b).

Professional Scepticism

Professional scepticism must be applied more rigorously because GTS has strong incentives to protect its sustainability branding. Compared with financial auditing, where bias often appears through earnings management or management override, the key ESG bias is reputation management or greenwashing. Sustainability assurance can improve credibility, but greenwashing concerns remain where corporate claims exceed verifiable evidence (International Federation of Accountants [IFAC], 2018; Free, Jones and Tremblay, 2024). KPMG should challenge the competence of Marketing / CSR personnel, the neutrality of the recycled-content methodology and the reasonableness of AI model assumptions. External allegations should be treated as contradictory evidence requiring follow-up. Overall, scepticism should be evidenced through documented challenges, alternative procedures and clear conclusions on whether remaining limitations are acceptable for limited assurance.

1.3 Materiality and Stakeholders

Materiality determines assurance focus and how misstatements or evidence limitations are

evaluated (IAASB, 2009c). In the financial audit, materiality is mainly assessed from financial statement users' perspective, using quantitative benchmarks and qualitative judgement. For GTS's ESG indicators, materiality is broader than in the financial audit because sustainability information may be non-financial, qualitative and relevant before any immediate financial statement effect arises. Under International Financial Reporting Standards (IFRS) S1, sustainability-related information is material if omitting, misstating or obscuring it could reasonably influence primary users' decisions (International Sustainability Standards Board [ISSB], 2023).

Double materiality is useful because it considers both how ESG issues affect GTS and how GTS affects people and the environment. Financial materiality covers greenwashing risk, carbon exposure, financing effects and supply-chain resilience, while impact materiality covers labour conditions, emissions and product-related environmental claims affecting broader stakeholders (European Financial Reporting Advisory Group [EFRAG], 2023). Where no simple financial threshold exists, KPMG should document severity, scale, scope, likelihood, stakeholder concern, regulatory exposure, media visibility and remediability. This aligns with GRI 3's focus on significant impacts on the economy, environment and people (GSSB, 2021).

Table 3. Stakeholder Needs and Assurance Focus for GTS

Stakeholder	Main information need	Assurance focus
Investors / lenders	Long-term risk and financing	Emissions, greenwashing, supplier risk
Customers	Credible green claims	Recycled content; carbon claims
Regulators	Non-misleading disclosure	Criteria, consistency, compliance
NGOs / media	Labour and environmental impact	Supplier X allegations
Employees	Workforce credibility	Diversity metrics; HR data

Note: HR = human resources.

Although stakeholder needs may affect the audit focus, as shown in Table 3, they

should not be equally determinative in every case. Investors and lenders are central to enterprise value, financing effects and transition risk (ISSB, 2023). NGOs, customers and employees point to other impact and credibility risks, such as responsible supply-chain conduct and workforce information (OECD, 2018; GSSB, 2016b). Where stakeholder interests conflict, KPMG should decide based on the agreed criteria, assurance scope and evidence gathered, rather than merely following the strongest voice (IAASB, 2021).

For GTS, Supplier X might not be quantitatively material to the financial statements, but it could still be material due to labour-rights, emissions, reputation and regulatory exposure. ISAE 3000 requires this judgement to be documented in materiality and risk assessment, using indicators such as NGO allegation severity, media attention, customer reliance on green claims, investor sensitivity to transition risk and possible financing effects (IAASB, 2013). Similarly, overstated recycled content may mislead customers and investors because GRI 301 addresses recycled input materials (GSSB, 2016a). Inaccurate Scope 1 and 2 GHG emissions could also distort climate-risk assessment under the GHG Protocol (Greenhouse Gas Protocol, 2015). Overall, GTS's materiality approach should be stakeholder-sensitive, impact-aware and supported by documented qualitative evidence, not reduced to financial thresholds alone.

Overall, Section 1 demonstrates that KPMG's ESG assurance work must be clearly bounded, evidence-driven and stakeholder-focused. Limited assurance is provided only on specific ESG indicators, not at audit level over the entirety of GTS's Sustainability Report. Hence, the team needs to define the scope, question weak ESG evidence, record materiality determinations and treat greenwashing allegations as triggers for targeted follow-up.

Section 2: Leveraging Technology for Enhanced Audit Efficiency and Effectiveness

In the GTS engagement, KPMG should use technology only where it improves evidence quality, traceability and audit efficiency without weakening professional judgement. Section 2 examines AI-supported procedures, blockchain verification and future auditor skills needed for responsible use (Stöckle, 2024).

2.1 AI in Audit Procedures

For GTS, AI-assisted procedures can improve coverage, precision and efficiency, but outputs should be validated rather than treated as audit evidence on their own. KPMG's audit technology discussion highlights AI's role in automation, data analysis, risk assessment and transaction-level analysis, while stressing that professional judgement and scepticism still apply to AI-generated outputs (Stöckle, 2024). The two most relevant applications are Natural Language Processing (NLP) for supplier evidence and predictive analytics for carbon-estimate testing. Table 4 links each application to the challenge addressed, efficiency gained and required control.

Table 4. AI Applications, Efficiencies and Controls for GTS

Challenge addressed	AI application	Efficiency gained	Risk and control
Supplier evidence reliability	NLP over contracts, emails and NGO reports	Wider review of unstructured evidence	Privacy limits; human review
Carbon model estimation	Predictive analytics and benchmarking	Independent reasonableness testing	Biased benchmarks; expert validation

NLP for Supplier Evidence

NLP might help assess weak evidence around Supplier X by scanning contracts, email survey responses, compliance certificates, NGO reports and media articles for red flags, inconsistencies or missing labour and emissions evidence. This improves efficiency because AI can process

large volumes of unstructured text and guide auditors towards higher-risk items (ACCA, 2019; IAASB, 2020). However, KPMG must consider consent, confidentiality, access rights, cross-border transfer restrictions and data minimisation. NLP may also misinterpret local terminology, translation or context. Hence, AI-flagged exceptions should be reviewed by auditors and, where needed, ESG or legal specialists.

Predictive Analytics for Carbon Estimates

Predictive analytics could be used to test the reasonableness of GTS's in-house AI model for product-level Scope 3 Category 11 carbon footprints. Scope 3 emissions are based on value-chain and customer-use assumptions and are difficult to quantify and validate directly (Greenhouse Gas Protocol, 2011). KPMG could develop an independent expectation based on product specifications, past energy-efficiency information, assumed user behaviour, production volumes and industry benchmarks. This would flag unusual outputs, overly optimistic carbon-reduction claims or estimates that appear inconsistent with comparable products, while allowing broader model testing than manual recalculation alone. However, KPMG's analytical model may also contain incomplete benchmarks, biased assumptions or model drift, so the team should assess benchmark quality, perform sensitivity analysis and involve carbon or data specialists where model risk is significant.

Controls over AI Reliance

The main audit risk is over-reliance. AI can improve anomaly detection and evidence coverage, but its outputs do not automatically constitute sufficient appropriate evidence under ISA 500 (IAASB, 2009b). To reduce black-box and bias risks, KPMG should use explainable outputs where possible, document tool validation, retain a mandatory human review threshold for all high-risk exceptions, and link each AI result to corroborating evidence. This is consistent with trustworthy AI principles such as validity, reliability, explainability, accountability and bias management (National Institute of Standards and Technology [NIST], 2023; OECD, 2024). Overall, AI should be used as a risk-targeting and evidence-enhancement tool, not as a substitute for professional judgement.

2.2 Blockchain for Trust and Verification

A blockchain-based supply-chain system would give KPMG shared, time-stamped and tamper-resistant ESG records. For GTS, this would reduce reliance on retrospective supplier email surveys and management summaries by providing a more continuous transaction trail (NIST, 2018; Chartered Professional Accountants of Canada [CPA Canada], 2018). However, blockchain strengthens the audit trail rather than replacing assurance procedures.

For recycled content, raw material suppliers could record certified recycled-material batches on the blockchain, including batch identity, certification details, weight and transfer records. These records could then be linked to production use and finished products, allowing KPMG to trace recycled materials from source to output and test whether the reported recycled content percentage is supported by a complete chain of custody (Institute of Chartered Accountants in England and Wales [ICAEW], 2017; NIST, 2018). The audit work would shift from checking selected supplier emails or certificates to testing whether blockchain records are complete, certificates are valid, transfers reconcile with production usage, and exceptions are investigated.

For Supplier X, blockchain technology could be used to record work-hour data, energy usage, emissions data, inspection results and compliance certificates to support ethical sourcing assurance. This would enable KPMG to spot missing uploads, late uploads, unusual working-hour patterns, abnormal energy use or certification gaps. It would also facilitate responsible supply-chain due diligence, considering that Supplier X is connected to labour and emissions allegations (OECD, 2018). However, this only works if Supplier X has the infrastructure, incentive and contractual requirement to participate. In Country X, cost, data privacy, weak systems and supplier resistance may limit adoption, so blockchain evidence will be reliable only if participation and data-entry responsibilities are enforceable.

The main implication for audit evidence quality is that blockchain may improve integrity,

traceability, timeliness and completeness, but it does not prove that source data are true. False certificates, biased supplier inputs or distorted sensor readings can still be recorded immutably; this is the "garbage in, garbage out" issue (CPA Canada, 2018). KPMG would therefore have to test supplier onboarding controls, access controls, permissions, data-input controls, smart contract logic, hashing or record-integrity controls, and exception logs before using blockchain evidence. Overall, blockchain could improve GTS's ESG assurance by making evidence more traceable and less dependent on management assertions, but its assurance value depends on source validation, supplier participation and strong system governance.

2.3 The Future Auditor—Skills and Ethics

The shift to ESG reporting and AI use changes auditors' competence profile. For new graduates entering ESG assurance, data analytics and AI literacy should be the most urgent skill set because auditors must understand data pipelines, model inputs, assumptions, bias risks and output limitations before using technology-enabled evidence. While automated tools can improve efficiency, outputs still require professional evaluation rather than automatic reliance (IAASB, 2020). This is critical for GTS, given its internal AI carbon model and AI-assisted supplier-evidence procedures. It also aligns with the increasing demand for auditors to understand how data analytics affects evidence evaluation (ICAEW, 2025).

The second skill set is ESG framework knowledge. Future auditors must understand how criteria such as ISAE 3000, ISAE 3410, GRI Standards, IFRS S1 and the GHG Protocol shape assurance work. For GTS, this knowledge helps KPMG assess whether recycled content, Scope 1 and 2 GHG emissions, workforce diversity metrics and carbon-related claims are prepared using suitable criteria. It also supports materiality judgement because sustainability information may affect users' decisions before any immediate financial statement impact arises (ISSB, 2023). More broadly, sustainability

reporting and assurance are becoming part of the expected competence profile for professional accountants (IFAC, 2025).

The third skill set is systems thinking combined with ethical judgement. ESG assurance requires auditors to connect supplier behaviour, production processes, product-use emissions, stakeholder trust and regulatory exposure. Supplier X's labour and emissions issues are therefore not isolated supplier matters; they affect GTS's sustainability claims, reputation and assurance risk. This reflects the broader nature of sustainability assurance, where evidence and judgement often extend across value chains and stakeholder impacts (IAASB, 2021). Systems thinking helps auditors trace these connections, while ethical judgement prevents AI tools from weakening accountability or professional scepticism, especially in supply-chain due diligence contexts (OECD, 2018).

A key ethical dilemma is AI bias and accountability. If KPMG uses an AI tool to scan supplier practices, incomplete training data or biased benchmarks may under-detect real labour issues or unfairly flag Supplier X. Trustworthy AI therefore requires attention to validity, reliability, explainability and bias management (NIST, 2023). This becomes harder where a vendor refuses to disclose training datasets, model logic or validation documentation. In that situation, KPMG should request transparency documentation, perform independent testing on sample outputs, involve ESG and data specialists, restrict reliance on opaque results, and document any unresolved limitation. These actions also reflect AI principles of transparency, human oversight and accountability (OECD, 2024). Overall, future auditor value depends on combining technological literacy with ESG judgement, systems thinking and ethical responsibility.

Taken together, Section 2 shows that AI and blockchain can improve GTS's ESG assurance only when supported by validation, source-data controls and human oversight. AI should be used to target risks and enhance evidence coverage, while blockchain can strengthen traceability. However, both technologies require skilled auditors who can

challenge outputs, manage ethical risks and preserve professional judgement.

Conclusion

This report has assessed the approach KPMG should take when conducting GTS's financial audit and concurrent limited assurance engagement over selected ESG indicators. The central conclusion is that ESG assurance should not be treated as a simple extension of the financial audit. While limited assurance can enhance credibility over selected ESG indicators, GTS's weak ESG data governance, recycled-content allegation, Supplier X concerns, carbon-estimation model and stakeholder-sensitive materiality create assurance risks that require targeted scepticism, clear boundaries and documented evidence work.

There are, however, some caveats to this analysis. The AI and blockchain applications discussed in this report are proposed for GTS and may not yet have been demonstrated at scale in limited assurance engagements. Procedures and reporting expectations may also evolve as future sustainability assurance requirements, such as the International Standard on Sustainability Assurance (ISSA) 5000, become more established (IAASB, 2024).

Within these limitations, the priority recommendations are:

1. **KPMG assurance work:** explain the assurance scope, manage the expectation gap and conduct targeted follow-up on recycled content, Supplier X and carbon-model assumptions.
2. **GTS governance:** involve finance and internal audit in ESG data collection, formalise recycled-content and emissions methodologies, and enhance supplier validation controls.

3. **Technology governance:** use AI and blockchain only where outputs are supported source-data controls, expert review, validation and human oversight.

Overall, assurance quality will depend on combining technology-enabled evidence with sceptical judgement and clear accountability.

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